# Integers

Positive and negative counting numbers with zero.

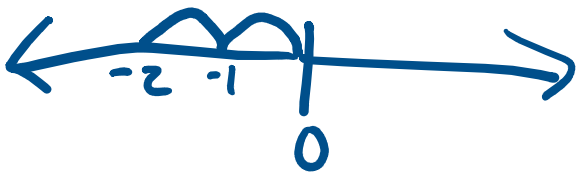
…-3, -2, -1, 0, 1, 2, 3…

Integers are discrete.

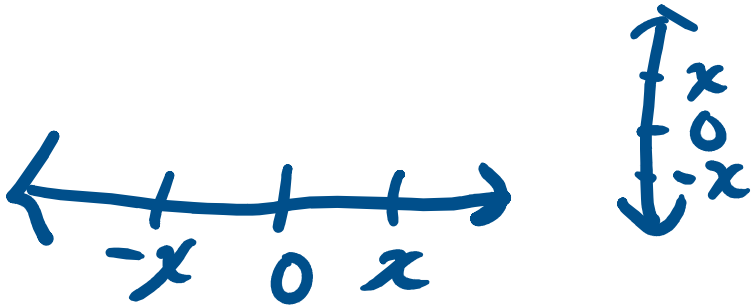
On a number line, you can think of the jumps going in the negative direction when you multiply by a negative number.

Remember, any negative number is a positive number going in the negative direction.

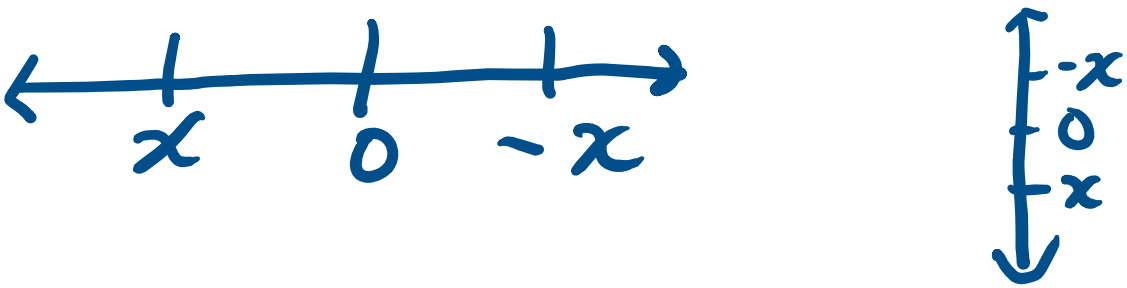
2*(-1) = -2



Can think of multiplying by -1 as flipping the number line. This is useful for inequalities.

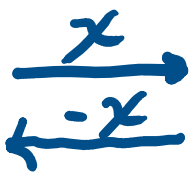



multiply by -1



Multiplying by -1 changes the direction of the vector

-x is -1*x  is (-1)x



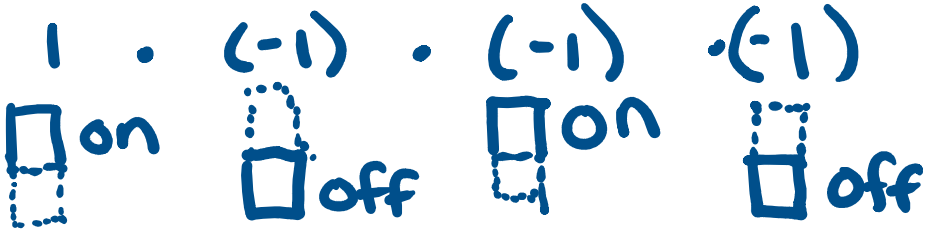
$$x(-1)(-1) = x$$

Multiplying by -1 changes the direction of the vector. You can model this with a light switch. If you start with on at positive one, then turning it off is multiplying it by negative one. If you flip the switch again, it is back up or on.

$1 \cdot (-1) \cdot (-1) \cdot (-1)$

If you start with on at positive one, then turning it off is multiplying it by negative one. If you flip the switch again, it is back up or on.

Do you see the pattern here?



If you start with a negative number, then multiplying by a negative number makes the answer positive.

We say, “a negative times a negative is a positive”.

$$(-1)\cdot(-1)$$

In multiplying two signed numbers, there are four possible situations:

$+1 \cdot +1 = +1$

$+1 \cdot -1 = -1$

$-1 \cdot +1 = -1$

$-1 \cdot -1 = +1$

order doesn't matter

Multiple multiplications of the same number is called exponentiation. We will come back to this later. but do you see that a negative number to an odd power is negative and a negative number to an even power is positive?

$(-1)^1 = (-1)$

$(-1)^2 = (-1)(-1) = +1$

$(-1)^3 = (-1)(-1)(-1) = -1$

$(-1)^4 = (-1)(-1)(-1)(-1) = +1$

$(-1)^{odd} = -1$

$(-1)^{even} = +1$

The same is true for division.

$$\frac{1}{1} = 1 \qquad \frac{-1}{-1} = 1$$

$$\frac{-1}{1} = -1 \qquad \frac{1}{-1} = -1$$

I never write the negative sign on the bottom because I lose it or forget about it!

$2(-x)$ is $-2x$

$-2(-x)$ is $2x$

# Division, fractions, and ratios can all be rates.

A rate is something per something else. Per means divided by.

Dollars per hour is an hourly salary.

Dollars per year

Miles per hour

Meters per second

The bottom or denominator is the units of the input and is for one of those units. So, dollars per hour is how many dollars you make in **one** hour.

# Factor-label method or unit conversion

In science, there are lots of formulas and you often need to convert between units of measurement. The units can cancel because all of the factors, other than the original are equal to one so the units cancel.

$$\frac{5\ \text{km}}{\text{hr}} \cdot \frac{1\ \text{hr}}{60\ \text{min}} \cdot \frac{1\ \text{min}}{60\ \text{sec}} \cdot \frac{1000\ \text{m}}{1\ \text{km}} = \frac{\text{m}}{\text{sec}}$$

In medicine, the dose for an antibiotic called amoxicillin is 40 mg per kg per day

amox $\frac{40\text{ mg}}{\text{Kg}}$

2.2 kg = 1 pound

1000 mg = 1 kg

$$\frac{x\ \text{lbs}}{1\ \text{baby}} \cdot \frac{2.2\ \text{kg}}{1\ \text{lbs}} \cdot \frac{1000\ \text{mg}}{\text{Kg}} = \frac{\text{mg}}{1\ \text{baby}}$$